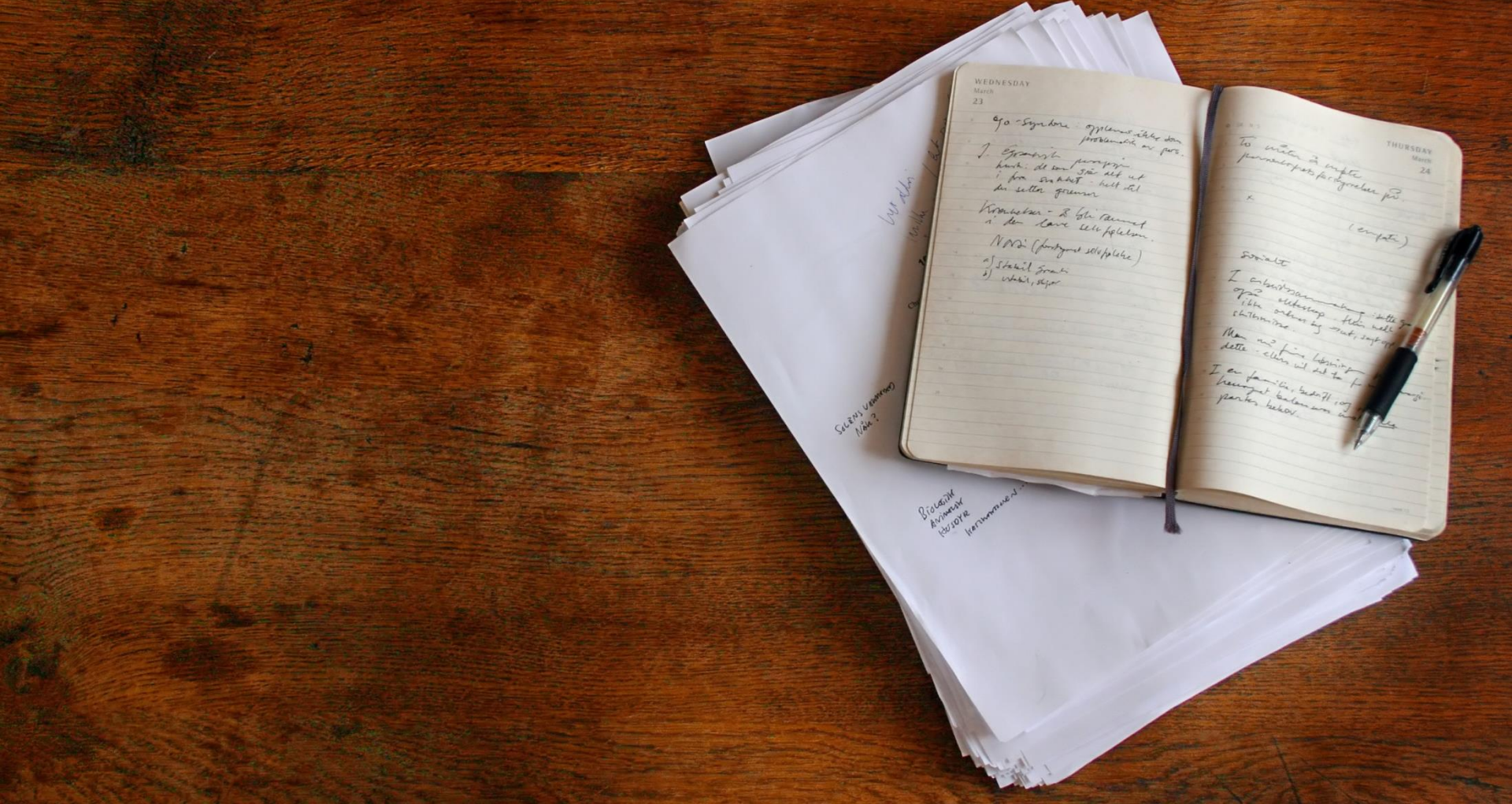




ACADEMIC WRITING (BETA) WEEK 2, LECTURE

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YOU NEVER GET
A SECOND
CHANCE
TO MAKE A
FIRST
IMPRESSION





<http://designtoimprovelife.dk/wp-content/uploads/2013/06/library-1024x691.jpg>
http://cx.aos.ask.com/question/aq/1400px-788px/business-solution_f4b4a1bcf89fb9d9.jpg

FINAL ASSIGNMENT

Write a 1000-word essay (plus or minus 10%) on a topic of your own choice, related to your academic discipline.

Use at least two to five academic articles, which are framed as a literature review-style text.

Your text should have 5 paragraphs of roughly the same length, structured as follows:

- An introductory paragraph with a thesis statement in the final sentence
- Three body paragraphs, each starting with a topic sentence
- A concluding paragraph

Title

Introduction paragraph

Body paragraph 1

Body paragraph 2

Body paragraph 3

Conclusion

List of references

Title

Introduction paragraph

Body paragraph 1

Body paragraph 2

Body paragraph 3

Conclusion

List of references

TODAY: INTRODUCTIONS

1. How do you start?
2. How do you end?
3. What goes in the middle?



HOW DO YOU START?



SAMPLE TEXT 1

Research in the area of smart environments has gained popularity over the last decade.

SAMPLE TEXT 2

The game of poker presents a serious challenge to artificial intelligence research.

From: Finnegan Southey, Michael Bowling, Bryce Larson, Carmelo Piccione, Neil Burch, Darse Billings, Chris Rayner. *Bayes' Bluff: Opponent Modelling in Poker*.

SAMPLE TEXT 3

As computers interact with us in increasingly complex and human ways through robots, wearable devices, PDAs, and various other ubiquitous interfaces, the psychological aspects of our relationships with them take on an increasingly important role [3].



HOW DO YOU END?



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We present a general Bayesian probabilistic model for *hold 'em* poker games, completely modelling the uncertainty in the game and the opponent.

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From: Bickmore & Schulman 2011. *A Virtual Laboratory for Studying Long-term Relationships between Humans and Virtual Agents*

SAMPLE TEXT 3

As computers interact with us in increasingly complex and human ways through robots, wearable devices, PDAs, and various other ubiquitous interfaces, the psychological aspects of our relationships with them take on an increasingly important role [3].

To address these needs, we have developed a “virtual laboratory” to support multiple, possibly concurrent, longitudinal studies of user interactions with a virtual agent. In the rest of this paper we discuss related work, and briefly describe the virtual laboratory system for conducting longitudinal studies of virtual agents in order to ground the subsequent discussion. We then describe a series of important issues that must be addressed when building systems for long-term use and longitudinal evaluation, using the virtual laboratory as an example. We then provide a brief primer on the statistical analysis of data from longitudinal studies and present preliminary results from the first study we have conducted using the virtual laboratory before concluding.

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FOR THE FINAL ASSIGNMENT:

- The thesis statement should be one sentence only.
- Three possible formats to choose from:
 - Thesis: There are three main reasons why teenagers use Facebook.
 - Statement of aims: This paper reviews/describes/examines why teenagers use Facebook.
 - Question: Why do teenagers use Facebook?
- No road maps (*I will first x, then y*).



WHAT GOES IN THE MIDDLE?

Using Fingerprint Authentication to Reduce System Security: An Empirical Study

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Abstract—Choosing the security architecture and policies for a system is a demanding task that must be informed by an understanding of user behavior. We investigate the hypothesis that adding visible security features to a system increases user confidence in the security of a system and thereby causes users to reduce how much effort they spend in other security areas. In our study, 96 volunteers each created a pair of accounts, one secured only by a password and one secured by both a password and a fingerprint reader. Our results strongly support our hypothesis—on average, when using the fingerprint reader, users created passwords that would take one three-thousandth as long to break, thereby *potentially* negating the advantage two-factor authentication could have offered.

Index Terms—user study, security policy, risk compensation, two-factor authentication

I. INTRODUCTION

Passwords remain the weakest component of many important security systems, so there is a concerted push from many directions to supplant or supplement passwords with less-fragile security measures. While this push has had some effects, particularly in environments that require more security, it has failed to replace passwords—the vast majority of computer users still use passwords on a day-to-day basis. Since the security of passwords relies so heavily on user behavior, studies that empirically examine patterns of password creation and use remain important in the evaluation of security policies.

Most empirical studies of user password behavior provide baseline information about how users create, remember, or use passwords [1]–[3] or the kind of passwords users create [4]. Some offer practical results based on comparing user groups given different sets of advice [5] or by persuading users to use stronger passwords [6]. A larger number of user studies have been done on alternatives to passwords, such as graphical passwords [7] or biometric authentication [8], though most of these have studied usability or subjective preferences rather than the security-related behavior of the users. Our study fills a void in the examination of a multi-factor authentication scheme by reviewing how users behave in such a system compared to how they behave in similar systems that use only single-factor authentication.

When policymakers decide to add to or improve upon password authentication rather than replace it with a different authentication scheme, the default assumption is that the additional layer or layers of security are essentially independent of other layers; that adding a security mechanism to a system

will not decrease the overall security of that system. From a software perspective, this assumption is generally true and can usually be verified. However, the security of a system is inextricably linked to and dependent upon the behavior of the authorized users of that system. If adding visible security features increases the level of confidence those users have in the security of the system, they may decide to expend less effort in behaving securely. This is an outcome predicted by risk compensation theory [9] or risk homeostasis theory [10], but has not been previously tested in the arena of computer security.

Our approach to partially address this question was to offer volunteers a monetary incentive to secure accounts they were told were going to be attacked. Each volunteer was asked to create two accounts with distinct passwords: one protected only by a password and one protected by both a password and a fingerprint reader—the most commonly used biometric for authentication. Users were given \$5 to divide as they saw fit between the two accounts and told that they would receive whatever money was in the accounts so long as that account was not compromised by the end of the study.

In addition to noting the evidential measure of security each user assigned to their accounts (the ratio of money put in each account), we asked the users directly about their confidence in the security measures protecting the two accounts. Finally, we asked some demographic questions in order to assess potential differences between user groups. We also observed users to see how many wrote down their passwords, but these data are unpaired to user accounts, so we cannot draw conclusions from it.

In order to analyze the results of our study we measured the strength of user passwords using a hybrid metric that attempts to estimate the number of guesses needed to guess each password. We used entropy measurements in this metric, but derived it primarily from the performance of several different password-cracking techniques on each password. Using this measure of strength, we found a large variance for how different each participant's pair of passwords was. However, across our entire study, the passwords connected to fingerprint-using accounts were $\frac{1}{2580}$ the strength of passwords for the password-only accounts. The decrease in strength was even more marked for several subgroups within our study, and we found strong correlations between a decrease in password strength and an expectation of the fingerprint reader providing

1 INTRODUCTION

With the growing acceptance that CO₂ emissions are a major cause for global warming and the changes of weather patterns, enterprises, governments, and society at large are beginning to consider environmental issues in the process of technology adoption leading to an increasing application of environmentally sound practices (Murugesan 2007). This includes the reduction of electric power consumption of IT hardware which uses up significant amounts of electricity, placing a heavy burden on the power grids (Murugesan 2008). According to a report provided by the Department of Energy (2007), data centers were estimated to have used 1.5 percent of all electricity in the United States in 2006, and their power demand is projected to grow 12 percent per year through 2011. Furthermore, many data centers are responsible for 30 to 40 percent of the energy consumption of an enterprise (O'Connor 2008). Therefore, a major objective of the IT industry is to reduce the power consumption and the environmental impact of IT to facilitate the emergence of a more sustainable environment.

Another major source of environmental problems is the production and disposal of IT, wherefore the green wave is sweeping the IT industry and the business domain. The trend of "greening" IT products, applications, services, and practices will continue since Green IT (also known as Green Computing) provides opportunities to reduce the accumulation of greenhouse gases in the atmosphere by reducing global CO₂ emissions. The current Green IT initiatives are especially reinforced by social and political pressure, governmental regulation, rising cost of waste disposal, corporate images, and public perception (Unhelkar and Dickens 2008, Kurp 2008).

Since little research has been conducted on Green IT and due to the several parallels between Green IT requirements and the characteristics of Grid technology, this article depicts the results of an analytical approach and outlines to what extent Grid technology can be used to implement the concepts of Green IT in order to leverage the inherent ecological and economical benefits. In addition, by applying the Resource-Based View to Green IT, the article illustrates that the assimilation of a Green Grid infrastructure leads to increased competitiveness of an enterprise. Due to the conceptual research approach, the guidelines proposed by Hirschheim (2008) have been considered to improve the rigor of the conducted research and to make a significant contribution to the knowledge of the relationship between Green IT, Grid technology, and competitive advantage of a firm.

The remainder of this article is organized as follows. First, a brief overview of Green IT and its concepts is provided in section 2. Section 3 emphasizes the need for Green IT infrastructures in order to reduce the power consumption in data centers and provides an introduction into Grid technology. Section 4 shows that the assimilation of Grid technology in the industry can be an effective implementation of Green IT concepts leading to increased competitiveness of enterprises, which will be illustrated in section 5. The article concludes with a summary of the key findings and an outlook to further research questions.

2 GREEN IT

Today, IT organisations are facing the dual challenge of providing more computing capabilities to meet dynamically changing and expanding business needs and thereby delivering such capabilities cost effectively as well as power and cooling efficient (Scaramella and Healey 2007). Due to this, IT manufacturers and vendors increasingly invest in Green IT initiatives and produce or use ecologically friendly and energy efficient IT solutions.

According to Murugesan (2008, pp. 25-26), Green IT is defined as "the study and practice of designing, manufacturing, using, and disposing of computers, servers, and associated subsystems (monitors, printers, storage devices, etc.) efficiently and effectively with minimal or no impact on the environment". This definition illustrates that the term "Green IT" is multifaceted and covers manifold

Using Artificial Team Members for Team Training in Virtual Environments

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Abstract. In a good team, members do not only perform their individual task, they also coordinate their actions with other members of the team. Developing such team skills usually involves exercises with all members playing their role. This approach is costly and has organizational and educational drawbacks. We developed a more efficient and flexible approach by setting training in virtual environments, and using intelligent software agents to play the role of team members. We developed a general framework for developing agents that, in a controlled fashion, execute the behavior that enables the human player (i.e., trainee) to effectively learn team skills. The framework is tested by developing and implementing various types of team agents in a game-based virtual environment.

Keywords: human-agent teams, virtual environments, team training

1. Introduction

A team of experts does not necessarily constitute an expert team. A team's performance is not only determined by how well the individual team members perform their tasks, but also by how well they practice their *team skills*. Not surprisingly, training team skills is an essential part of the education of many professions demanding a tight collaboration [10]. For example, fire fighters not only learn how to operate a fire hose (task behavior), but also how they must communicate relevant information about the fire's progress to their colleagues (team behavior).

Usually, team skills are trained by exercises with a (potentially large) number of participants, each playing a dedicated role in a given scenario. Such large scale team exercises have a number of drawbacks. First, this type of team training requires ample resources (both human and financial) and is difficult to organize. Furthermore, it occurs frequently that not all members of a team are available for training at the same time and location, and that role players (e.g. staff personnel) need to take their place. It can also be that the team members vary in competence, thus having different training needs (e.g. a team consisting of novice and advanced trainees). For example, an urban patrol scenario contains many roles which are either not relevant for soldiers practicing their team skills (e.g. *local inhabitant*), or for which some soldiers holding that rank have already completed their training.

We propose to approach this problem by combining recent developments in virtual environments (VEs) and intelligent software agents. We aim at developing a game-based platform for team training in which some team members are played by humans, and some are played by software agents. In this way, team training can be made less costly, as not all roles have to be fulfilled by humans anymore. Furthermore, training can be better tailored to the specific learning objectives of a specific individual, as we can control more precisely the team behavior of artificial team members than that of human team members (which may be trainees themselves).

Whereas computer-controlled characters in VEs are common in computer games, these characters are usually programmed with scripts which list a number of trigger/event rules. The type of behavior required for teamwork is too complex to be programmed in these scripting languages. Therefore, we have developed a framework for developing software agents whose behavior in a VE is based on concepts as goals, plans, and beliefs, and which is "aware" of its team.

The behavior of the software agent is directed by a teamwork model. Teamwork models have been studied for several decades by the AI community, e.g. [12]. Typical applications of these models are mixed human-agent teams [11], where computers assist humans in a flexible way, i.e. as a teammate. Our application to team training is novel, as our prime interest is not to develop a "perfect" team member which is always subjected to humans. Rather, we aim at developing a realistic team member which exhibits natural behavior and from which humans can *learn*. This makes human trainees better team players "in the real world" after they have completed their training. For example, a real team member sometimes forgets to communicate an important piece of information, for instance if he is too busy with other tasks. Implementing this property into an artificial team members allow us to design realistic experiences, from which the trainee can learn how to deal with such team task problems.

Our research objectives can thus be formulated as follows. Firstly, we aim at developing generic teamwork models tailored to training fundamental team skills. Secondly, we aim to implement these models to develop an agent-based artificial team member, situated in a virtual environment. This paper describes the concepts and design of our agent-based team training approach, followed by a proof-of-concept study, showing that it is a feasible alternative to human-based team training. We demonstrate our approach by presenting *Samurai*, a software agent that in a virtual environment performs a team task in a coordinated fashion with a human player (trainee).

The paper is organized as follows. The following section describes how teamwork is implemented in our system. Section 3 describes the implementation of *Samurai* and the integration of *Samurai* with the virtual environment. Section 4 gives the training case and event traces. A conclusion is given in Section 5.

2. Training of Teamwork

Teamwork is a topic of great complexity and breadth and no consensus has yet been reached on an exact definition. Most simply, a team is a group of people working together to achieve a common goal. This requires team members to maintain *common ground*, to be *mutually predictable*, and to be *mutually directable* [6]. Related

SAMPLE TEXT 1

Research in the area of smart environments has gained popularity over the last decade.

Most attention has been directed towards health monitoring and activity recognition [1-4]. Recently, assistive health care systems have started making an impact in society, especially in countries where human care-giving facilities are expensive and a large population of adults prefers an independent lifestyle.

There are currently 18 million people worldwide who are diagnosed with dementia and this number is predicted to reach 35 million by 2050 [6]. These older adults face problems completing both simple (e.g. eating, dressing) and complex (e.g. cooking, taking medicine) Activities of Daily Living (ADLs) [7].

An automated computerized system that would be able to provide some of the facilities of a human caregiver is the call of the hour and would help in alleviating the burden of many caregivers that are helping a large section of the population.

In this chapter, we describe in detail the problem of delivering in-home interventions and compare some of the prospective approaches.

FOR THE FINAL ASSIGNMENT:

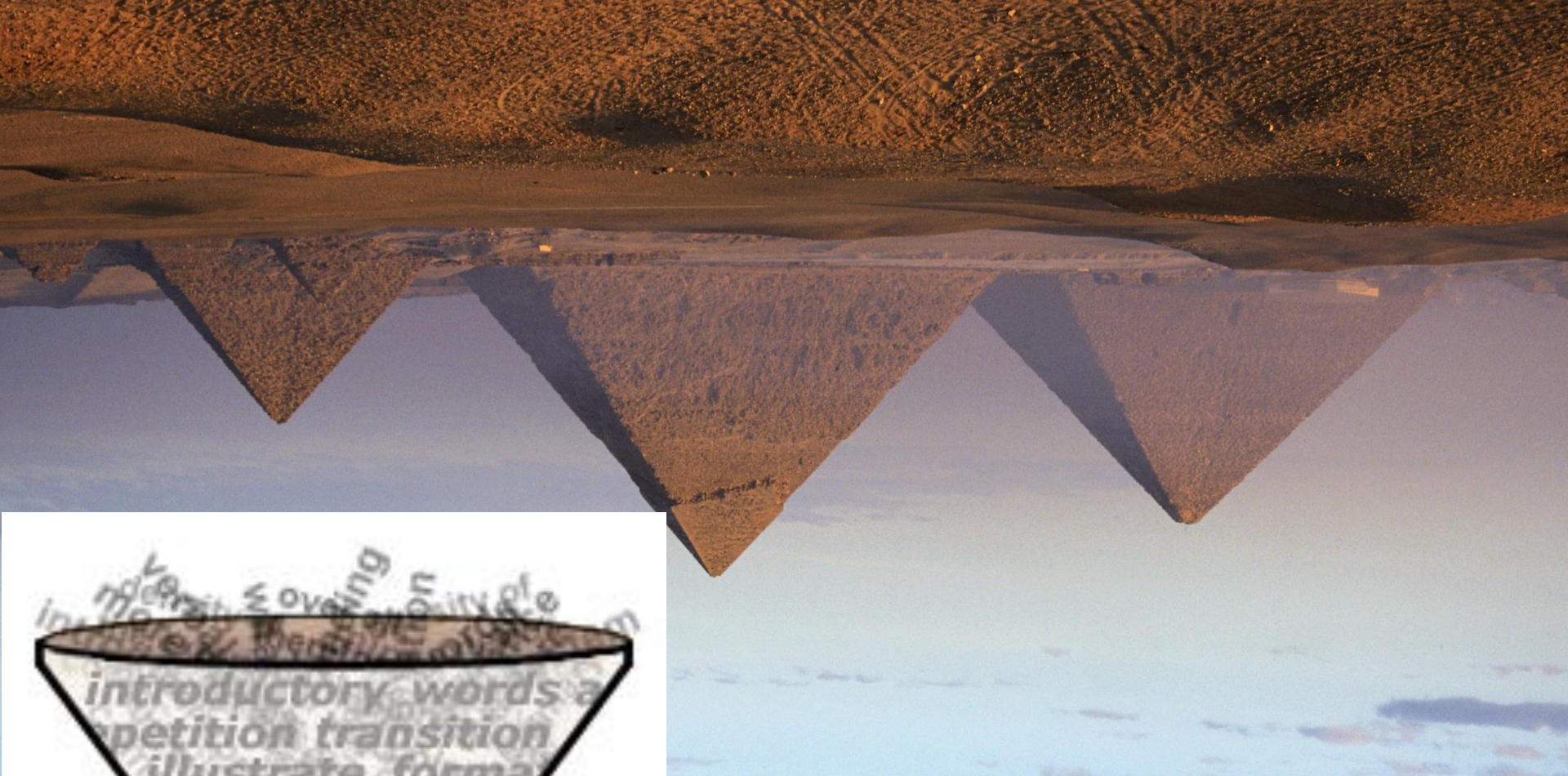
- Introduction of just one paragraph (roughly 200 words)

209 words, 9 sentences:

Over the last two decades, the Open Source Software (OSS) community has steadily amassed many thousands of developers and has seen the rise of numerous organizations, adopting its philosophy of making software available to the world – free of charge (Spinellis & Vaggelis, 2011). Recognition of their work is often the sole reward, for all the spare time they have dedicated to the development and distribution of software – often for the benefit of others. Moreover, anybody may study the source code, make modifications, and redistribute the software as they see fit. Many alternatives, to commercial software packages, have seen the light in this manner, providing noncommercial users with software that was previously unaffordable. OSS's underlying user licenses also allow commercial organizations to make use of these alternatives. Well-known examples of these substitutes are the Apache web server, the MySQL database server and the Linux Operating System. Despite the obvious cost benefit to using OSS, there remains the perception that many businesses are not willing to deploy these alternatives in their ICT-infrastructure. Determining the actual success of OSS, within commercial organizations, has been the subject of various articles and research over the years. Unfortunately, the findings are frequently contradictory, and may have even exacerbated that uncertainty (Ven, Verhelst & Mannaert, 2008).



SO, WHAT ABOUT THAT
PYRAMID?



http://www.brunswick.k12.me.us/hdwyer/files/2011/09/thesis_funnel1.png

<http://cdn.history.com/sites/2/2014/01/pyramids-giza.jpg>

TODAY: INTRODUCTIONS

1. How do you start?
2. How do you end?
3. What goes in the middle?

This is the basic format for assignment 1 (more about that in the seminar)

Join at menti.com | use code 5258 9366

What are the functions of the first sentence?

17 responses

Introducing to what the paper will be about.

Hook

establish the area it wants to cover

Intrigue the reader

To get the readers attention

Hooking readers into reading the full paper

To capture the reader.

Describe to the reader the basic premise

To hook the reader

Attracting reader to the text



AM



Menti

BETA lecture 2



Choose a slide to present

What are the functions of the first sentence?
17 responses

Why is the thesis statement the most important sentence of your text?
0 responses

QUIZTIME

Take a pen and paper, or your laptop, and write down the answers to the following questions:

1. What are the two functions of the first sentence?
2. Why is the thesis statement the most important sentence of your text?
3. What does the funnel/inverted pyramid shape mean?

QUIZTIME - ANSWERS

Take a pen and paper, or your laptop, and write down the answers to the following questions:

1. What are the two functions of the first sentence?

A first sentence should draw the reader's attention and should introduce the main area of the topic of the paper.

2. Why is the thesis statement the most important sentence of your text?

It contains your main aim with the text. If your aim is not clear, a reader doesn't know what to expect and won't make the decision to read on.

3. What does the funnel/inverted pyramid shape mean?

You start with more general information and end with the most specific topic (your thesis statement) > that is the real topic of your paper.

COMING UP NEXT...

1. This week's seminar:
 - Practising with introductions
 - Diagnostic grammar exercises
2. Friday 11 April: deadline for assignment 1
3. Next week:
 - Quoting and referring